NT-1b Vector (Gauge Boson) Form Factors: Literature Conventions and Benchmarks

Literature Review — Independent Cross-Verification

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Abstract

This document collects all operator data, sign conventions, trace identities, and benchmark values needed to compute the one-loop nonlocal form factors $h_C^{(1)}(\Box/m^2)$ and $h_R^{(1)}(\Box/m^2)$ for a **vector gauge boson** (spin-1) on a curved background, using the Barvinsky–Vilkovisky / Codello–Zanusso heat kernel formalism. All formulas are extracted from the primary literature; no original derivations are performed (those belong to the derivation stage). Each value is annotated with its provenance: [exact from lit.] = extracted verbatim from a paper equation; [verified] = cross-checked against ≥ 2 independent sources; [inferred by analogy] = inferred by analogy with the scalar (Phase 1) computation.

This is the literature review for Phase 2 (vector), which will be critically reviewed before proceeding to the derivation stage.

Key differences from Phase 1 (scalar):

1. The vector endomorphism $\mathbf{U} = \mathrm{Ric}$ is **matrix-valued** (not proportional to R).
2. The bundle curvature $\Omega_{\mu\nu} \neq 0$ (Riemann tensor).
3. Faddeev–Popov ghosts (2 real scalar fields) must be subtracted.
4. The CZ form factor $\mathrm{tr}[\mathbf{U}\, f_{\mathbf{U}}\, \mathbf{U}] = R_{\mu\nu}\, f_{\mathbf{U}}\, R^{\mu\nu}$ contributes to **both** C^2 and R^2 sectors (unlike the scalar case).

Key references:

- Codello–Zanusso (CZ): arXiv:1203.2034 [1].
- Teixeira–Shapiro–Ribeiro (TSR): arXiv:2003.04503 [2].
- Vassilevich: Phys. Rept. **388** (2003) 279 [hep-th/0306138] [3].
- Barvinsky–Vilkovisky (BV): Nucl. Phys. B **333** (1990) 471 [4].
- Codello–Percacci–Rahmede (CPR): arXiv:0805.2909 [5].

Critical result: $\beta_W^{(1)} = 1/10$, **not** 13/120 as stated in `constants.py` (line 81). The erroneous value 13/120 arises from subtracting only one ghost instead of two (ghost + anti-ghost). See Section 3 and Benchmark B1.

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1 Vector Field Operator Data

1.1 Classical action and kinetic operator

The action for a massless Abelian vector field A_μ on a d-dimensional Riemannian manifold (M, g):

$$S[A] = \frac{1}{4}\int \mathrm{d}^d x\,\sqrt{g}\,F_{\mu\nu}F^{\mu\nu}\,, \qquad F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu\,. \tag{1}$$

In Feynman gauge ($\alpha = 1$), the gauge-fixed kinetic operator is

$$\Delta_{\text{vector}} = -\Box\,\delta^\alpha{}_\beta + R^\alpha{}_\beta\,, \tag{2}$$

where $R^\alpha{}_\beta$ is the Ricci tensor (mixed components). [exact from lit.] [Vassilevich eq. (1.4.3); CPR eq. (A.4)]

Remark 1.1 (Weitzenbock identity). The operator (2) is precisely the **Hodge–de Rham Laplacian** on 1-forms:

$$\Delta_{\text{HdR}} = \mathrm{d}\delta + \delta\mathrm{d} = -\nabla^2 + \mathrm{Ric}\,, \tag{3}$$

where d is the exterior derivative and δ its formal adjoint. The Weitzenbock identity establishes this equality. [exact from lit.] [Vassilevich Section 1.4; Jost, "Riemannian Geometry and Geometric Analysis", Thm. 4.3.1]

1.2 Identification of operator data

The operator (2) is of **Laplace type**, acting on sections of the cotangent bundle T^*M.

Convention 1.2 (CZ convention)**.** With $\Delta = -D^2 + \mathbf{U}$ [CZ eq. (2)]:

$$\boxed{\mathbf{U}_{\text{CZ}} = R^\alpha{}_\beta = \mathrm{Ric} \qquad \text{(the Ricci tensor as an endomorphism of } T^*M\text{).}} \tag{4}$$

[verified] [CPR eq. (A.4): "$E^{(\mathrm{M})} = \mathrm{Ricci}$"; Vassilevich eq. (1.4.3)]

Convention 1.3 (Our convention — matching NT-1)**.** [our convention] With $\Delta = -D^2 - E$ (NT-1 Convention 2.6):

$$\boxed{E = -\mathbf{U}_{\text{CZ}} = -R^\alpha{}_\beta = -\mathrm{Ric}\,.} \tag{5}$$

Check: $-\Box - E = -\Box + \mathrm{Ric}$, recovering (2). ✓

Convention 1.4 (BV convention)**.** The BV potential $P_{\text{BV}} = -\mathbf{U} + R/6 = E + R/6$:

$$\boxed{(P_{\text{BV}})^\alpha{}_\beta = -R^\alpha{}_\beta + \frac{R}{6}\,\delta^\alpha{}_\beta = \mathrm{Ric}\cdot(-1) + \frac{R}{6}\,\mathrm{Id}\,.} \tag{6}$$

[verified] [TSR eq. (3), CZ Appendix A]
Key structural property: P_{BV} is **not proportional to the identity** for the vector field. This is a major difference from the scalar case (where $P_{\text{BV}} = -\tilde{\xi}\,R$) and the Dirac case (where $P_{\text{BV}} = -R/12$).

1.3 Bundle curvature

The connection on T^*M is the Levi-Civita connection. The bundle curvature is [Vassilevich eq. (1.4.2)]:

$$\boxed{(\Omega_{\mu\nu})^\alpha{}_\beta = [D_\mu, D_\nu]^\alpha{}_\beta = R^\alpha{}_{\beta\mu\nu}\,,} \tag{7}$$

where $R^\alpha{}_{\beta\mu\nu}$ is the Riemann curvature tensor. [verified] [Vassilevich eq. (1.4.2); CPR Appendix A]

Remark 1.5 (Contrast with scalar and Dirac). For the scalar, $\Omega_{\mu\nu} = 0$ (trivial bundle). For the Dirac spinor, $\Omega_{\mu\nu} = (1/4)R_{\rho\sigma\mu\nu}\gamma^\rho\gamma^\sigma$ (spin curvature). For the vector, $\Omega_{\mu\nu} =$ Riemann directly.

1.4 Complete operator data summary

Quantity	Value (vector)	Provenance	Source
$\mathrm{tr}(\mathrm{Id})$	4	[exact from lit.]	$\dim T^*M = 4$
$\Omega_{\mu\nu}$	$R^\alpha{}_{\beta\mu\nu}$	[verified]	Vassilevich eq. (1.4.2)
E (our convention)	$-R^\alpha{}_\beta$	[verified]	eq. (5)
$\mathbf{U}_{\mathrm{CZ}} = -E$	$R^\alpha{}_\beta$	[verified]	CPR eq. (A.4)
$(P_{\mathrm{BV}})^\alpha{}_\beta$	$-R^\alpha{}_\beta + \frac{R}{6}\delta^\alpha{}_\beta$	[verified]	eq. (6)

Remark 1.6 (No non-minimal coupling parameter). Unlike the scalar field, the vector field has **no free coupling parameter** ξ. The curvature coupling $R^\alpha{}_\beta$ is uniquely fixed by gauge invariance (Weitzenbock identity). This means $h_C^{(1)}$ and $h_R^{(1)}$ are parameter-free functions of x only.

2 Trace Identities for the Vector Bundle

All traces below are over the **fiber indices** (the 4-vector bundle T^*M in $d = 4$).

2.1 Elementary traces

$$\boxed{\mathrm{tr}(\mathrm{Id}) = 4\,, \qquad \mathrm{tr}(E) = -R\,, \qquad \mathrm{tr}(\mathbf{U}) = R\,.} \tag{8}$$

[verified] The first is $\dim T^*M = d = 4$. The second: $\mathrm{tr}(-R^\alpha{}_\beta) = -R^\alpha{}_\alpha = -R$. The third: $\mathrm{tr}(\mathrm{Ric}) = R$ by definition of scalar curvature.

2.2 BV potential traces

$$\mathrm{tr}(P_{\mathrm{BV}}) = \mathrm{tr}\Big(-\mathrm{Ric} + \frac{R}{6}\mathrm{Id}\Big) = -R + \frac{4R}{6} = -\frac{R}{3}\,, \tag{9}$$

$$\mathrm{tr}(\mathbf{U} + \tfrac{R}{6}\mathrm{Id}) = \mathrm{tr}\Big(\mathrm{Ric} + \frac{R}{6}\mathrm{Id}\Big) = R + \frac{2R}{3} = \frac{5R}{3}\,. \tag{10}$$

[our convention] Note: the BV potential trace is $-R/3$ in our (Vassilevich) convention, but the CZ-convention "hat-P" has trace $5R/3$. In the CZ formalism we use $\hat{P}_{\rm CZ} = \mathbf{U} + R/6 \cdot \mathrm{Id} = \mathrm{Ric} + (R/6)\mathrm{Id}$, which enters the heat kernel expansion directly. **We work with the CZ convention throughout this document.**

2.3 Quadratic traces

Proposition 2.1 (Quadratic fiber traces for the vector field)**.**

$$\mathrm{tr}(E^2) = \mathrm{tr}(\mathrm{Ric}^2) = R_{\mu\nu}R^{\mu\nu}\,, \tag{11}$$

$$\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu}) = -R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\,, \tag{12}$$

$$\mathrm{tr}\Big[(\mathbf{U} + \tfrac{R}{6}\mathrm{Id})\, f_{\mathbf{U}}(\Box)\, (\mathbf{U} + \tfrac{R}{6}\mathrm{Id})\Big] = R_{\mu\nu}\, f_{\mathbf{U}}(\Box)\, R^{\mu\nu} + \frac{4}{9}\, R\, f_{\mathbf{U}}(\Box)\, R\,. \tag{13}$$

Proof of (12). [Vassilevich Section 4.1]
$\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu}) = R^{\alpha}{}_{\beta\mu\nu}\, R^{\beta}{}_{\alpha}{}^{\mu\nu}$. Using the first-pair antisymmetry $R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}$:

$$R^{\alpha}{}_{\beta\mu\nu}\, R^{\beta}{}_{\alpha}{}^{\mu\nu} = g^{\alpha\gamma}\, R_{\gamma\beta\mu\nu}\, g^{\beta\delta}\, R_{\delta\alpha}{}^{\mu\nu} = R_{\alpha\beta\mu\nu}\, R^{\beta\alpha\mu\nu} = -R_{\alpha\beta\mu\nu}\, R^{\alpha\beta\mu\nu} = -R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\,.$$

□

Proof of (13). Denoting $\hat{P}_{\rm CZ} = \mathrm{Ric} + (R/6)\mathrm{Id}$, we compute $\hat{P} f_{\mathbf{U}} \hat{P}$ component by component:

$$\begin{aligned}
\Big[\hat{P}\, f_{\mathbf{U}}\, \hat{P}\Big]^{\alpha}{}_{\alpha} &= \Big(R^{\alpha}{}_{\beta} + \tfrac{R}{6}\delta^{\alpha}{}_{\beta}\Big) f_{\mathbf{U}}(\Box) \Big(R^{\beta}{}_{\alpha} + \tfrac{R}{6}\delta^{\beta}{}_{\alpha}\Big) \\
&= R^{\alpha}{}_{\beta}\, f_{\mathbf{U}}\, R^{\beta}{}_{\alpha} + \frac{R}{6}\, f_{\mathbf{U}}\, R^{\alpha}{}_{\alpha} + R^{\alpha}{}_{\alpha}\, f_{\mathbf{U}}\, \frac{R}{6} + \frac{R^2}{36} \cdot 4 \\
&= R_{\mu\nu}\, f_{\mathbf{U}}\, R^{\mu\nu} + \frac{R}{6}\, f_{\mathbf{U}}\, R + \frac{R}{6}\, f_{\mathbf{U}}\, R + \frac{R^2}{9} \\
&= R_{\mu\nu}\, f_{\mathbf{U}}\, R^{\mu\nu} + \frac{R}{3}\, f_{\mathbf{U}}\, R + \frac{R}{9}\, f_{\mathbf{U}}\, R\,,
\end{aligned}$$

where in the last step we used $f_{\mathbf{U}}(0) = 1/2$ to evaluate the constant $(R^2/9) \cdot f_{\mathbf{U}}$—but actually at the nonlocal level the $R^2/36 \cdot 4$ term gives $(4/36)R\, f\, R = (1/9)R\, f\, R$ when we note that R is a scalar and $f_{\mathbf{U}}$ acts trivially on it (the factor $R/9$ arises from $R/6 \cdot R/6 \cdot \mathrm{tr}(\mathrm{Id})$ with the $f_{\mathbf{U}}$ acting on the second $R/6$, but since $R/6$ is a scalar, $f_{\mathbf{U}}(\Box)(R/6)$ just gives $(R/6) \cdot f_{\mathbf{U}}$ evaluated on R).

More carefully: at $\mathcal{O}(\mathcal{R}^2)$ on a flat background, $f_{\mathbf{U}}(\Box)$ acts on each curvature factor independently. The result is:

$$\mathrm{tr}[\hat{P}\, f_{\mathbf{U}}\, \hat{P}] = R_{\mu\nu}\, f_{\mathbf{U}}(\Box)\, R^{\mu\nu} + \frac{4}{9}\, R\, f_{\mathbf{U}}(\Box)\, R\,.$$

where $4/9 = 1/3 + 1/9 = (3+1)/9$. □

Remark 2.2 (Critical structural difference from scalar/Dirac). For the scalar ($\hat{P}_{\rm CZ} = (\xi + 1/6)R$) and Dirac ($\hat{P}_{\rm CZ} = R/4 + R/6 = 5R/12$) cases, $\hat{P}$ is **proportional to the identity**, so $\mathrm{tr}[\hat{P}\, f_{\mathbf{U}}\, \hat{P}] = (\mathrm{tr}\, \hat{P})^2/N \cdot f_{\mathbf{U}} \propto R\, f_{\mathbf{U}}\, R$ (pure R^2 structure). For the vector, $\hat{P}$ has nontrivial matrix structure (Ric is not proportional to Id on a generic manifold), so the $f_{\mathbf{U}}$ term contributes to **both** the C^2 and R^2 sectors. This is the key complication in the vector computation.

3 Faddeev–Popov Ghost Subtraction

3.1 Ghost operator

The Faddeev–Popov procedure for a gauge field in Feynman gauge introduces a **complex** scalar ghost field c and anti-ghost $\bar{c}$, or equivalently **two real** scalar ghost fields. Each ghost has the kinetic operator

$$\Delta_{\text{ghost}} = -\Box\,, \tag{14}$$

i.e. a minimally coupled scalar ($\xi = 0$, $m = 0$). [exact from lit.] [CPR eq. (A.4): "$E^{(\text{gh})} = 0$"; Vassilevich Section 1.4]

Convention 3.1 (Ghost operator data)**.** For each real ghost scalar:

$$\text{tr}(\text{Id})_{\text{ghost}} = 1\,, \quad E_{\text{ghost}} = 0\,, \quad \mathbf{U}_{\text{ghost}} = 0\,, \quad \Omega_{\text{ghost}} = 0\,. \tag{15}$$

The BV potential: $P_{\text{BV,ghost}} = R/6$. This is a **minimally coupled massless scalar**. [verified] [CPR eq. (A.4)]

3.2 Physical gauge field effective action

The one-loop effective action for the **physical** gauge field is:

$$\boxed{\Gamma^{(1)}_{\text{gauge}} = \frac{1}{2}\,\text{Tr}\ln\Delta_{\text{vector}} - \text{Tr}\ln\Delta_{\text{ghost}}\,,} \tag{16}$$

where $-\,\text{Tr}\ln\Delta_{\text{ghost}}$ accounts for **both** the ghost and anti-ghost (a complex scalar contributes $-\,\text{Tr}\ln\Delta$, or equivalently $2\times(-\frac{1}{2}\,\text{Tr}\ln\Delta)$ for two real scalars with the wrong-sign statistics). [exact from lit.] [Vassilevich eq. (1.4.4); Peskin–Schroeder Chapter 16]

In terms of the form factors:

$$\boxed{h^{(1)}_{C,R}(x) = h^{(1,\text{unconstr})}_{C,R}(x) - 2\,h^{(0)}_{C,R}(x;\xi=0)\,,} \tag{17}$$

where $h^{(1,\text{unconstr})}$ is the unconstrained vector contribution and $h^{(0)}(\xi = 0)$ is the minimally coupled scalar form factor from Phase 1. [verified] [CPR Section 3.3; BV (1990)]

Remark 3.2 (Why 2 ghosts, not 1). The error in `constants.py` (which lists "Expected: 13/120") arises from subtracting only **one** ghost: $14/120 - 1/120 = 13/120$. The correct subtraction uses **two** real scalar ghosts: $14/120 - 2 \times 1/120 = 12/120 = 1/10$. The Faddeev–Popov procedure **always** produces a complex ghost field (ghost + anti-ghost = 2 real d.o.f.). [Weinberg, QFT Vol. II, Section 15.6; Peskin–Schroeder Section 16.2]

4 Sign Convention Comparison Table

Quantity	**CZ (1203.2034)**	**CPR (0805.2909)**	**Our NT-1**
Vector field (gauge boson, Feynman gauge):			
Endomorphism	$\mathbf{U} = \mathrm{Ric}$	$E^{(\mathrm{M})} = \mathrm{Ric}$	$E = -\mathrm{Ric}$
Bundle curvature	$\Omega_{\mu\nu} = R^{\alpha}{}_{\beta\mu\nu}$	same	$\Omega_{\mu\nu} = R^{\alpha}{}_{\beta\mu\nu}$
BV potential	$\hat{P} = \mathrm{Ric} + \frac{R}{6}\mathrm{Id}$	—	$P_{\mathrm{BV}} = -\mathrm{Ric} + \frac{R}{6}\mathrm{Id}$
Ghost (minimally coupled scalar):			
Endomorphism	$\mathbf{U} = 0$	$E^{(\mathrm{gh})} = 0$	$E = 0$

[verified] All entries cross-checked. The key convention is the **same** as Phase 1: $E_{\mathrm{ours}} = -\mathbf{U}_{\mathrm{CZ}}$. We work with CZ form factors throughout, using $\mathbf{U} = +\mathrm{Ric}$.

5 Known β-Coefficients (Benchmarks B1–B2)

The one-loop effective action for the physical gauge field in the Weyl–R^2 basis:

$$\Gamma^{(1)}_{\mathrm{gauge}} = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{g}\left\{h_C^{(1)}(\Box/m^2)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + h_R^{(1)}(\Box/m^2)\,R^2\right\} + \mathrm{E}_4 \text{ terms}\,. \tag{18}$$

The local limits define the β-coefficients: $\beta_W^{(1)} = h_C^{(1)}(0)$ and $\beta_R^{(1)} = h_R^{(1)}(0)$.

5.1 Weyl-squared coefficient

Benchmark 5.1 ($\beta_W^{(1)}$ for gauge boson — B1).

$$\boxed{\beta_W^{(1)} = \frac{1}{10} = \frac{12}{120}\,.} \tag{19}$$

[verified] Derived from **four** independent paths:

Source 1 (CPR direct): CPR [5] eq. (matterergeII) gives the C^2 matter contribution per gauge field: $36/180 = 1/5$. In the $\frac{1}{2}\,\mathrm{Tr}\ln$ normalization: $\beta_W = (1/2)\times 1/5 = 1/10$. [exact from lit.] [CPR eq. (matterergeII) with $n_M = 1$]

Source 2 (BV table): From the β_W table in NT-1 literature conventions (`NT1_literature_conventions` line 826): "Gauge vector: 1/10" citing BV (1990). [exact from lit.] [NT-1 master table]

Source 3 (CZ form factor assembly): $h_C^{(1,\mathrm{unconstr})}(0) = 2\cdot f_{\mathrm{Ric}}(0) + \frac{1}{2}f_{\mathbf{U}}(0) - 2\cdot f_{\Omega}(0) = 2\cdot\frac{1}{60} + \frac{1}{2}\cdot\frac{1}{2} - 2\cdot\frac{1}{12} = \frac{1}{30} + \frac{1}{4} - \frac{1}{6} = \frac{7}{60}$. Ghost: $h_C^{(0,\xi=0)}(0) = \frac{1}{120}$. Physical: $\frac{7}{60} - 2\cdot\frac{1}{120} = \frac{7}{60} - \frac{1}{60} = \frac{6}{60} = \frac{1}{10}$. [CZ eq. (5.4) values, assembled with ghost subtraction]

Source 4 (Seeley–DeWitt a_4 direct): From Vassilevich, the a_4 coefficient for the unconstrained vector field in the $\{C^2, R^2, E_4\}$ basis gives C^2 coefficient $= 42/360 = 7/60$. Ghost: $3/360 = 1/120$ each. Physical: $7/60 - 2/120 = 14/120 - 2/120 = 12/120 = 1/10$. [Vassilevich eq. (4.1), specialized to vector]

5.2 R^2 coefficient

Benchmark 5.2 ($\beta_R^{(1)}$ for gauge boson — B2).

$$\boxed{\beta_R^{(1)} = 0\,.} \tag{20}$$

[verified] Derived from **three** independent arguments:

Source 1 (CPR direct): CPR [5] eq. (matterergeII) gives the R^2 matter contribution: $(1/180) \times 5\, n_S$. Crucially, **only scalars** contribute to R^2 — the n_D and n_M terms have zero coefficient. For $n_M = 1$ (gauge field): R^2 coefficient $= 0$. [exact from lit.] [CPR eq. (matterergeII), R^2 line]

Source 2 (Conformal invariance): The Maxwell action $F_{\mu\nu}F^{\mu\nu}$ is **Weyl invariant** in $d = 4$. An R^2 counterterm would break Weyl invariance, so it cannot be generated. Only the Weyl-invariant C^2 counterterm is generated. [Standard argument; analogous to Dirac $\beta_R^{(1/2)} = 0$]

Source 3 (CZ form factor assembly): $h_R^{(1,\text{unconstr})}(0) = 1/36$ and $h_R^{(0,\xi=0)}(0) = 1/72$. Physical: $1/36 - 2 \times 1/72 = 1/36 - 1/36 = 0$. [CZ eq. (5.4) values, assembled]

Remark 5.3 (Pattern of conformal invariance). The vanishing β_R for both the massless Dirac fermion and the gauge boson reflects conformal invariance of the respective classical actions in $d = 4$. For the scalar, $\beta_R^{(0)} = 0$ only at conformal coupling $\xi = 1/6$. This pattern is:

Field	β_W	β_R
Scalar (general ξ)	$1/120$	$(1/2)(\xi - 1/6)^2$
Dirac	$-1/20$	0
Gauge vector	$1/10$	0

[verified] [CPR Table I; Vassilevich Table 1 (implicitly)]

6 CZ Form Factor Formulas Applied to the Vector

6.1 Master function (same as Phase 1)

$$\varphi(x) \equiv f(x) = \int_0^1 \mathrm{d}\alpha\, e^{-\alpha(1-\alpha)\,x}\,. \tag{21}$$

[exact from lit.] [CZ eq. (5.3)]

Taylor expansion and asymptotics (same as Phase 1):

$$\varphi(x) = 1 - \frac{x}{6} + \frac{x^2}{60} - \frac{x^3}{840} + O(x^4)\,, \tag{22}$$

$$\varphi(x) = \frac{2}{x} + \frac{4}{x^2} + O(x^{-3}) \quad \text{as } x \to \infty\,. \tag{23}$$

6.2 Five CZ form factors (universal)

The five CZ form factors (identical to Phase 1, independent of spin):

$$f_{\mathrm{Ric}}(x) = \frac{1}{6x} + \frac{\varphi(x) - 1}{x^2}\,, \tag{24}$$

$$f_R(x) = \frac{\varphi}{32} + \frac{\varphi}{8x} - \frac{7}{48x} - \frac{\varphi - 1}{8x^2}\,, \tag{25}$$

$$f_{R\mathbf{U}}(x) = -\frac{\varphi}{4} - \frac{\varphi - 1}{2x}\,, \tag{26}$$

$$f_{\mathbf{U}}(x) = \frac{\varphi}{2}\,, \tag{27}$$

$$f_\Omega(x) = -\frac{\varphi - 1}{2x}\,. \tag{28}$$

[exact from lit.] [CZ eq. (5.3); confirmed from arXiv LaTeX source of 1203.2034]

Local limits:

$$\boxed{f_{\mathrm{Ric}}(0) = \frac{1}{60}\,, \quad f_R(0) = \frac{1}{120}\,, \quad f_{R\mathbf{U}}(0) = -\frac{1}{6}\,, \quad f_{\mathbf{U}}(0) = \frac{1}{2}\,, \quad f_\Omega(0) = \frac{1}{12}\,.} \tag{29}$$

[verified] [CZ eq. (5.4)]

6.3 Application to the unconstrained vector field

The CZ heat kernel expansion (eq. (5.1) of [1]) at $\mathcal{O}(\mathcal{R}^2)$ gives, after taking the fiber trace:

$$\begin{aligned}\Gamma^{(1,\mathrm{unconstr})}\Big|_{\mathcal{R}^2} = \frac{1}{16\pi^2}\int \mathrm{d}^4x\,\sqrt{g}\,\Big\{&\underbrace{4\,R_{\mu\nu}\,f_{\mathrm{Ric}}\,R^{\mu\nu}}_{\text{Term 1}} + \underbrace{4\,R\,f_R\,R}_{\text{Term 2}} \\ &+ \underbrace{R\,f_{R\mathbf{U}}\,R}_{\text{Term 3}} + \underbrace{R_{\mu\nu}\,f_{\mathbf{U}}\,R^{\mu\nu} + \frac{4}{9}\,R\,f_{\mathbf{U}}\,R}_{\text{Term 4}} \\ &+ \underbrace{\Big(-R_{\alpha\beta\mu\nu}\,f_\Omega\,R^{\alpha\beta\mu\nu}\Big)}_{\text{Term 5}}\Big\}\,.\end{aligned} \tag{30}$$

The five terms arise from:

Term	CZ structure	Vector trace
1	$\mathrm{tr}[\mathrm{Id}\cdot R_{\mu\nu}\,f_{\mathrm{Ric}}\,R^{\mu\nu}]$	$4\cdot R_{\mu\nu}\,f_{\mathrm{Ric}}\,R^{\mu\nu}$
2	$\mathrm{tr}[\mathrm{Id}\cdot R\,f_R\,R]$	$4\cdot R\,f_R\,R$
3	$\mathrm{tr}[R\,f_{R\mathbf{U}}\,\mathbf{U}]$	$R\,f_{R\mathbf{U}}\cdot\mathrm{tr}(\mathrm{Ric}) = R\,f_{R\mathbf{U}}\,R$
4	$\mathrm{tr}[\mathbf{U}\,f_{\mathbf{U}}\,\mathbf{U}]$	$R_{\mu\nu}\,f_{\mathbf{U}}\,R^{\mu\nu}$ (Prop. 2.1)
	$\mathrm{tr}[(\frac{R}{6}\mathrm{Id})\,f_{\mathbf{U}}\,(\mathrm{Ric}+\frac{R}{6}\mathrm{Id})]$	$+\frac{4}{9}\,R\,f_{\mathbf{U}}\,R$ (cross + diagonal terms)
5	$\mathrm{tr}[\Omega_{\mu\nu}\,f_\Omega\,\Omega^{\mu\nu}]$	$-R_{\alpha\beta\mu\nu}\,f_\Omega\,R^{\alpha\beta\mu\nu}$

Remark 6.1 (Term 3 simplification). For the term $\mathrm{tr}[R\, f_{R\mathbf{U}}\, \mathbf{U}]$: since R is a scalar, it passes through the fiber trace, giving $R \cdot f_{R\mathbf{U}} \cdot \mathrm{tr}(\mathbf{U}) = R \cdot f_{R\mathbf{U}} \cdot R$. At $\mathcal{O}(\mathcal{R}^2)$, $f_{R\mathbf{U}}(\Box)$ acts on $\mathrm{tr}(\mathbf{U}) = R$ as a scalar function, so the trace commutes with $f_{R\mathbf{U}}(\Box)$ up to $\mathcal{O}(\mathcal{R}^3)$ corrections (since the form factor is computed on a flat background).

Remark 6.2 (Term 4 decomposition). [SIGN CORRECTED] *(Corrected by Literature Review.)* In the CZ expansion (eq. 5.1 of [1]), Term 4 comes from $\mathrm{tr}[\mathbf{U}\, f_{\mathbf{U}}\, \mathbf{U}]$ (using the **unshifted** endomorphism $\mathbf{U}$, not $\hat{P}_{\mathrm{CZ}} = \mathbf{U} + (R/6)\mathrm{Id}$). For the vector: $\mathrm{tr}[\mathrm{Ric}\, f_{\mathbf{U}}\, \mathrm{Ric}] = R_{\mu\nu}\, f_{\mathbf{U}}\, R^{\mu\nu}$, which splits into $(1/2)\, f_{\mathbf{U}}\, C^2 + (1/3)\, f_{\mathbf{U}}\, R^2$ via the Weyl decomposition of $R_{\mu\nu}R^{\mu\nu}$. The $R/6$ cross-terms are already captured by $f_{R\mathbf{U}}$ and f_R in the CZ five-form-factor structure.

7 Weyl-Basis Form Factors

7.1 Gauss–Bonnet identity and Weyl decomposition

The same identities as Phase 1 apply. At $\mathcal{O}(\mathcal{R}^2)$ in $d = 4$, dropping topological E_4 terms:

$$R_{\mu\nu}R^{\mu\nu} = \frac{1}{2}\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{1}{3}\, R^2\,, \tag{31}$$

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 2\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \frac{1}{3}\, R^2\,. \tag{32}$$

[exact from lit.] [CZ Appendix A; standard GR identity]

7.2 Unconstrained vector form factors

Collecting (30) into the $\{R_{\mu\nu}R^{\mu\nu},\, R^2,\, R_{\alpha\beta\mu\nu}R^{\alpha\beta\mu\nu}\}$ basis:

$$\text{Coeff. of } R_{\mu\nu}R^{\mu\nu}: \quad 4\, f_{\mathrm{Ric}} + f_{\mathbf{U}}\,, \tag{33}$$

$$\text{Coeff. of } R^2: \quad 4\, f_R + f_{R\mathbf{U}}\,, \tag{34}$$

$$\text{Coeff. of } R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}: \quad -f_\Omega\,. \tag{35}$$

Applying the Weyl decomposition (31)–(32):

Proposition 7.1 (Unconstrained vector Weyl-basis form factors)**.**

$$\boxed{h_C^{(1,\mathrm{unconstr})}(x) = 2\, f_{\mathrm{Ric}}(x) + \frac{1}{2}\, f_{\mathbf{U}}(x) - 2\, f_\Omega(x)\,,} \tag{36}$$

$$\boxed{h_R^{(1,\mathrm{unconstr})}(x) = \frac{4}{3}\, f_{\mathrm{Ric}}(x) + 4\, f_R(x) + f_{R\mathbf{U}}(x) + \frac{1}{3}\, f_{\mathbf{U}}(x) - \frac{1}{3}\, f_\Omega(x)\,.} \tag{37}$$

Proof. From the coefficients (33)–(35) and the decomposition (31)–(32):

C^2 coefficient: $(4f_{\mathrm{Ric}} + f_{\mathbf{U}}) \cdot \frac{1}{2} + (-f_\Omega) \cdot 2 = 2f_{\mathrm{Ric}} + \frac{1}{2}f_{\mathbf{U}} - 2f_\Omega$.

R^2 coefficient: $(4f_{\mathrm{Ric}} + f_{\mathbf{U}}) \cdot \frac{1}{3} + (4f_R + f_{R\mathbf{U}}) + (-f_\Omega) \cdot \frac{1}{3} = \frac{4}{3}f_{\mathrm{Ric}} + \frac{1}{3}f_{\mathbf{U}} + 4f_R + f_{R\mathbf{U}} - \frac{1}{3}f_\Omega$.

Note (corrected by Literature Review): The CZ expansion (eq. 5.1 of [1]) uses $\mathrm{tr}[\mathbf{U}\, f_{\mathbf{U}}\, \mathbf{U}]$ (not $\mathrm{tr}[\hat{P}\, f_{\mathbf{U}}\, \hat{P}]$); the $R/6$ shift is already distributed across $f_{R\mathbf{U}}$ and f_R. Hence $\mathrm{tr}[\mathrm{Ric}\, f_{\mathbf{U}}\, \mathrm{Ric}] = R_{\mu\nu}R^{\mu\nu}\, f_{\mathbf{U}}$, which contributes $(1/3)\, f_{\mathbf{U}}$ to the R^2 sector via the Weyl decomposition $R_{\mu\nu}R^{\mu\nu} \to (1/3)\, R^2$. □

7.3 Local-limit verification (unconstrained)

At $x = 0$, using (29):

$$\begin{aligned} h_C^{(1,\text{unconstr})}(0) &= 2 \cdot \frac{1}{60} + \frac{1}{2} \cdot \frac{1}{2} - 2 \cdot \frac{1}{12} = \frac{1}{30} + \frac{1}{4} - \frac{1}{6} \\ &= \frac{2 + 15 - 10}{60} = \frac{7}{60}\,. \end{aligned} \tag{38}$$

$$h_R^{(1,\text{unconstr})}(0) = \frac{4}{3} \cdot \frac{1}{60} + 4 \cdot \frac{1}{120} + \left(-\frac{1}{6}\right) + \frac{1}{3} \cdot \frac{1}{2} - \frac{1}{3} \cdot \frac{1}{12}\,. \tag{39}$$

Computing with common denominator 180: $= \frac{4}{180} + \frac{6}{180} - \frac{30}{180} + \frac{30}{180} - \frac{5}{180} = \frac{4+6-30+30-5}{180} = \frac{5}{180} = \frac{1}{36}$.

$$\boxed{h_C^{(1,\text{unconstr})}(0) = \frac{7}{60} = \frac{14}{120}\,, \qquad h_R^{(1,\text{unconstr})}(0) = \frac{1}{36}\,.} \tag{40}$$

7.4 Physical gauge field form factors

Applying the ghost subtraction (17) with the Phase 1 scalar results:

$$h_C^{(0)}(x; \xi = 0) = \frac{1}{2} f_{\text{Ric}}(x) = \frac{1}{12x} + \frac{\varphi - 1}{2x^2}\,, \tag{41}$$

$$h_R^{(0)}(x; \xi = 0) = f_{R,\text{bis}}(x) = \frac{1}{3} f_{\text{Ric}}(x) + f_R(x)\,. \tag{42}$$

[verified] [Phase 1 scalar document, eqs. (41) and (42)]

Local limits for ghosts: $h_C^{(0)}(0; \xi = 0) = 1/120$ and $h_R^{(0)}(0; \xi = 0) = 1/72$. [verified] [Phase 1 Benchmarks B5 and B6]

7.5 Closed-form results

Proposition 7.2 (Physical gauge field form factors — closed form)**.**

$$\boxed{h_C^{(1)}(x) = \frac{\varphi(x)}{4} + \frac{6\varphi(x) - 5}{6x} + \frac{\varphi(x) - 1}{x^2}\,,} \tag{43}$$

$$\boxed{h_R^{(1)}(x) = -\frac{\varphi(x)}{48} + \frac{11 - 6\varphi(x)}{72x} + \frac{5\big(\varphi(x) - 1\big)}{12x^2}\,.} \tag{44}$$

Proof. We compute $h^{(1)} = h^{(1,\text{unconstr})} - 2\,h^{(0)}(\xi = 0)$ step by step.
$h_C^{(1)}$**:** From (36): $h_C^{(1,\text{unconstr})} = 2 f_{\text{Ric}} + \frac{1}{2} f_{\mathbf{U}} - 2 f_\Omega$.

Substituting the explicit CZ form factors:

$$\begin{aligned} 2 f_{\text{Ric}} &= \frac{1}{3x} + \frac{2(\varphi - 1)}{x^2}\,, \\ \frac{1}{2} f_{\mathbf{U}} &= \frac{\varphi}{4}\,, \\ -2 f_\Omega &= \frac{\varphi - 1}{x}\,. \end{aligned}$$

Sum: $h_C^{(1,\text{unconstr})} = \frac{\varphi}{4} + \frac{1}{3x} + \frac{\varphi-1}{x} + \frac{2(\varphi-1)}{x^2} = \frac{\varphi}{4} + \frac{3\varphi-2}{3x} + \frac{2(\varphi-1)}{x^2}$.

Ghost subtraction: $2\,h_C^{(0)}(\xi=0) = \frac{1}{6x} + \frac{\varphi-1}{x^2}$.

Difference:

$$\begin{aligned} h_C^{(1)} &= \frac{\varphi}{4} + \frac{3\varphi-2}{3x} - \frac{1}{6x} + \frac{2(\varphi-1)}{x^2} - \frac{\varphi-1}{x^2} \\ &= \frac{\varphi}{4} + \frac{2(3\varphi-2)-1}{6x} + \frac{\varphi-1}{x^2} \\ &= \frac{\varphi}{4} + \frac{6\varphi-5}{6x} + \frac{\varphi-1}{x^2}\,. \end{aligned}$$

$h_R^{(1)}$**:** From the corrected (37): $h_R^{(1,\text{unconstr})} = \frac{4}{3}f_{\text{Ric}} + 4f_R + f_{R\mathbf{U}} + \frac{1}{3}f_{\mathbf{U}} - \frac{1}{3}f_\Omega$.

Ghost subtraction: $2\,h_R^{(0)}(\xi=0) = 2\,f_{R,\text{bis}} = \frac{2}{3}f_{\text{Ric}} + 2\,f_R$, so:

$$h_R^{(1)} = \frac{2}{3}f_{\text{Ric}} + 2f_R + f_{R\mathbf{U}} + \frac{1}{3}f_{\mathbf{U}} - \frac{1}{3}f_\Omega\,.$$

Substituting the explicit CZ form factors:

$$\begin{aligned} \frac{2}{3}f_{\text{Ric}} &= \frac{1}{9x} + \frac{2(\varphi-1)}{3x^2}\,, \\ 2f_R &= \frac{\varphi}{16} + \frac{\varphi}{4x} - \frac{7}{24x} - \frac{\varphi-1}{4x^2}\,, \\ f_{R\mathbf{U}} &= -\frac{\varphi}{4} - \frac{\varphi-1}{2x}\,, \\ \frac{1}{3}f_{\mathbf{U}} &= \frac{\varphi}{6}\,, \\ -\frac{1}{3}f_\Omega &= \frac{\varphi-1}{6x}\,. \end{aligned}$$

Constants (no $1/x$): $\frac{\varphi}{16} - \frac{\varphi}{4} + \frac{\varphi}{6} = \varphi\Big(\frac{3-12+8}{48}\Big) = -\frac{\varphi}{48}$.

$1/x$ terms: $\frac{1}{9x} + \frac{\varphi}{4x} - \frac{7}{24x} - \frac{\varphi-1}{2x} + \frac{\varphi-1}{6x} = \frac{1}{9x} + \frac{\varphi}{4x} - \frac{7}{24x} - \frac{\varphi}{2x} + \frac{1}{2x} + \frac{\varphi}{6x} - \frac{1}{6x}$
$= \frac{1}{9x} - \frac{7}{24x} + \frac{1}{2x} - \frac{1}{6x} + \varphi\Big(\frac{1}{4} - \frac{1}{2} + \frac{1}{6}\Big)\frac{1}{x}$
$= \frac{1}{9x} - \frac{7}{24x} + \frac{1}{3x} + \varphi\cdot\frac{3-6+2}{12x} = \frac{1}{9x} - \frac{7}{24x} + \frac{1}{3x} - \frac{\varphi}{12x}$.

Common denominator $72x$: $\frac{8-21+24}{72x} - \frac{\varphi}{12x} = \frac{11}{72x} - \frac{6\varphi}{72x} = \frac{11-6\varphi}{72x}$.

$1/x^2$ terms: $\frac{2(\varphi-1)}{3x^2} - \frac{\varphi-1}{4x^2} = \frac{8(\varphi-1)-3(\varphi-1)}{12x^2} = \frac{5(\varphi-1)}{12x^2}$.

Combining all terms:

$$h_R^{(1)}(x) = -\frac{\varphi}{48} + \frac{11-6\varphi}{72x} + \frac{5(\varphi-1)}{12x^2}\,.$$

□

8 Verification of Local Limits

8.1 Benchmark B1: $h_C^{(1)}(0) = 1/10$

From (43) with $\varphi = 1 - x/6 + x^2/60 + \cdots$:

$(6\varphi-5)/(6x) = (1 - x + x^2/10 + \cdots)/(6x) = 1/(6x) - 1/6 + x/60 + \cdots$

$(\varphi - 1)/x^2 = (-x/6 + x^2/60 + \cdots)/x^2 = -1/(6x) + 1/60 + \cdots$
Poles cancel: $1/(6x) - 1/(6x) = 0$. ✓
Constant: $\varphi(0)/4 + (-1/6) + 1/60 = 1/4 - 1/6 + 1/60 = (15 - 10 + 1)/60 = 6/60 = 1/10$.

$$\boxed{h_C^{(1)}(0) = \beta_W^{(1)} = \frac{1}{10}\,.} \tag{45}$$

8.2 Benchmark B2: $h_R^{(1)}(0) = 0$

From (44):
$(11 - 6\varphi)/(72x) = (5 + x - x^2/10 + \cdots)/(72x) = 5/(72x) + 1/72 + \cdots$
$5(\varphi - 1)/(12x^2) = 5(-x/6 + x^2/60 + \cdots)/(12x^2) = -5/(72x) + 5/720 + \cdots$
Poles cancel: $5/(72x) - 5/(72x) = 0$. ✓
Constant: $-1/48 + 1/72 + 1/144 = (-3 + 2 + 1)/144 = 0/144 = 0$.

$$\boxed{h_R^{(1)}(0) = \beta_R^{(1)} = 0\,.} \tag{46}$$

8.3 Benchmark B3: $h_C^{(1,\text{unconstr})}(0) = 7/60$

Already computed in (38).

8.4 Benchmark B4: $h_R^{(1,\text{unconstr})}(0) = 1/36$

Already computed following (39).

9 Taylor Expansions and Asymptotics (Benchmarks B5–B8)

9.1 Taylor expansion at $x = 0$

Benchmark 9.1 (Taylor coefficients — B7, B8).

$$h_C^{(1)}(x) = \frac{1}{10} - \frac{11}{420}\,x + O(x^2)\,, \tag{47}$$

$$h_R^{(1)}(x) = \frac{1}{630}\,x + O(x^2)\,. \tag{48}$$

Derivation of Taylor coefficients. Using $\varphi(x) = 1 - x/6 + x^2/60 + \cdots$:
$h_C^{(1)}$**:** $\varphi/4 = 1/4 - x/24 + \cdots$
$(6\varphi - 5)/(6x) = (1 - x + \cdots)/(6x) = 1/(6x) - 1/6 + x/60 + \cdots$
$(\varphi - 1)/x^2 = -1/(6x) + 1/60 - x/840 + \cdots$
Sum (poles cancel): $h_C^{(1)} = [1/4 - 1/6 + 1/60] + [-1/24 + 1/60 - 1/840]x + \cdots$ $= 1/10 + (-35 + 14 - 1)x/840 + \cdots = 1/10 - 22x/840 + \cdots = 1/10 - 11x/420 + \cdots$
$h_R^{(1)}$**:** Since $h_R^{(1)}(0) = 0$, the expansion starts at $O(x)$.
$-\varphi/48 = -1/48 + x/288 + \cdots$
$(11 - 6\varphi)/(72x) = (5 + x + \cdots)/(72x) = 5/(72x) + 1/72 - x/720 + \cdots$
$5(\varphi - 1)/(12x^2) = -5/(72x) + 1/144 - x/2016 + \cdots$

Sum (poles cancel): $h_R^{(1)} = [-1/48 + 1/72 + 1/144] + [1/288 - 1/720 - 1/2016]x + \cdots = 0 + (35 - 14 - 5)x/10080 + \cdots = 16x/10080 + \cdots = x/630 + \cdots$ □

Remark 9.2 (Physical interpretation of $h_R^{(1)} = O(x)$). The vanishing of $h_R^{(1)}(0)$ means the R^2 sector has **no logarithmic divergence** for the gauge field. The leading nonlocal correction is proportional to $x = -\Box/m^2$, corresponding to a finite $R\Box R$ contribution to the effective action.

9.2 Large-x asymptotics

Benchmark 9.3 (Large-x behavior — B5, B6). Using $\varphi(x) \to 2/x + 4/x^2 + O(x^{-3})$:

$$h_C^{(1)}(x) = -\frac{1}{3x} + \frac{1}{x^2} + O(x^{-3}), \tag{49}$$

$$h_R^{(1)}(x) = \frac{1}{9x} - \frac{7}{12x^2} + O(x^{-3}). \tag{50}$$

Derivation of asymptotics. $h_C^{(1)}$**:** $\varphi/4 \to 1/(2x)$; $(6\varphi - 5)/(6x) \to (12/x - 5)/(6x) = -5/(6x) + 2/x^2$; $(\varphi - 1)/x^2 \to -1/x^2 + 2/x^3$. Sum: $h_C^{(1)} \to 1/(2x) - 5/(6x) + 2/x^2 - 1/x^2 = (3-5)/(6x) + 1/x^2 = -1/(3x) + 1/x^2$.
$h_R^{(1)}$**:** $-\varphi/48 \to -1/(24x)$; $(11 - 6\varphi)/(72x) \to (11 - 12/x)/(72x) = 11/(72x) - 1/(6x^2)$; $5(\varphi - 1)/(12x^2) \to -5/(12x^2)$. Sum: $-1/(24x) + 11/(72x) + (-1/6 - 5/12)/x^2 = (-3 + 11)/(72x) + (-2 - 5)/(12x^2) = 8/(72x) - 7/(12x^2) = 1/(9x) - 7/(12x^2)$. □

10 Structural Benchmarks (B9–B12)

Benchmark 10.1 (Ghost subtraction — B9). The physical gauge field form factors are obtained by subtracting **two** real scalar ghost contributions:

$$h_{C,R}^{(1)}(x) = h_{C,R}^{(1,\text{unconstr})}(x) - 2\, h_{C,R}^{(0)}(x; \xi = 0). \tag{51}$$

At the local limit: $14/120 - 2 \times 1/120 = 12/120 = 1/10$ for β_W, and $1/36 - 2 \times 1/72 = 0$ for β_R. [verified]

Benchmark 10.2 (No ξ-dependence — B10). The vector form factors $h_C^{(1)}(x)$ and $h_R^{(1)}(x)$ have **no free coupling parameter**. The curvature coupling $\mathbf{U} = \mathrm{Ric}$ is uniquely fixed by gauge invariance. This contrasts with the scalar case where $h_R^{(0)}(x; \xi)$ depends quadratically on ξ.

Benchmark 10.3 (Conformal invariance $\Rightarrow \beta_R = 0$ — B11). The Maxwell action is Weyl-invariant in $d = 4$: $F_{\mu\nu}F^{\mu\nu}$ is invariant under $g_{\mu\nu} \to e^{2\sigma} g_{\mu\nu}$. An R^2 counterterm would break Weyl invariance, so $\beta_R^{(1)} = 0$. This is the vector analogue of the Dirac result $\beta_R^{(1/2)} = 0$. [verified] [Standard conformal field theory argument; CPR confirms numerically]

Benchmark 10.4 (Gauss–Bonnet coefficient — B12). The E_4 coefficient of the a_4 Seeley–DeWitt coefficient for the physical gauge field is:

$$\alpha_{E_4}^{(1)} = -\frac{31}{180}. \tag{52}$$

[verified] [CPR eq. (matterergell) with $n_M = 1$: $-62/360 = -31/180$ in the $(1/2)\,\mathrm{Tr}\ln$ normalization (matching the β_W convention used throughout this document)]

Note: E_4 is topological in $d = 4$ and does not contribute to equations of motion. This benchmark serves as a cross-check on the full a_4 decomposition (involving ALL three curvature invariants, not just C^2 and R^2).

11 Comparison Across Spins

	Scalar ($s = 0$, $\xi = 0$)	**Dirac** ($s = 1/2$)	**Vector** ($s = 1$)
tr(Id)	1	4	4
U	ξR	$R/4$	Ric
$\Omega_{\mu\nu}$	0	spin curvature	Riemann
β_W	$1/120$	$-1/20$	$1/10$
β_R	$1/72$	0	0
$h_C(0)/h_C^{(0)}(0)$	1	-6	12
α_{E_4}	$-1/360$	$11/360$	$-31/180$
Conformal?	at $\xi = 1/6$	yes	yes

Remark 11.1 (Sign pattern of β_W). The signs alternate: scalar (+), Dirac (−), vector (+). This reflects the statistics: bosonic fields contribute positively to β_W, while fermionic fields contribute negatively. The relative magnitudes $1 : 6 : 12$ (for $|\beta_W|/\beta_W^{(0)}$) encode both the multiplicity (tr(Id)) and the enhancement from bundle curvature.

Remark 11.2 (Dirac sign convention). The Dirac $\beta_W^{(1/2)} = -1/20$ includes the fermionic sign from $\Gamma^{(1)}_{\text{Dirac}} = -\operatorname{Tr}\ln\Delta_{\text{Dirac}}$ (the extra minus sign for fermions). The "absolute" form factor value is $|h_C^{(1/2)}(0)| = 1/20$, but the physical β_W carries the negative sign.

12 Error in `constants.py` (CORRECTED)

The file `analysis/sct_tools/constants.py`, line 81, contains:

```
1: None,   # vector -- to be computed in NT-1b Phase 2
           # Expected: 13/120 (CPR Table 1 / Vassilevich)
```

The value 13/120 is WRONG. The correct value is:

$$\beta_W^{(1)} = \frac{1}{10} = \frac{12}{120}\,. \tag{53}$$

Root cause: The error comes from subtracting only **one** ghost (1/120) from the unconstrained vector (14/120), giving $14/120 - 1/120 = 13/120$. The correct subtraction requires **two** ghosts (ghost + anti-ghost): $14/120 - 2/120 = 12/120 = 1/10$.
Also: The attribution "CPR Table 1" is misleading. CPR = Codello–Percacci–Rahmede [5] (arXiv:0805.2909), not Codello–Percacci–Rachwal. The CPR paper does not contain a Table 1 with the value 13/120; their matter contribution formula (eq. matterergeII) gives $36/180 = 1/5$ per gauge field (in the full-trace normalization), which yields $\beta_W = 1/10$ in the half-trace convention.

13 Summary: All 12 Benchmarks

The following 12 benchmarks define the complete set of tests for the Phase 2 derivation of the vector (gauge boson) nonlocal form factors:

B1: $\beta_W^{(1)} = h_C^{(1)}(0) = \frac{1}{10}$ [verified] (4 sources)

B2: $\beta_R^{(1)} = h_R^{(1)}(0) = 0$ [verified] (3 sources)

B3: $h_C^{(1,\text{unconstr})}(0) = \frac{7}{60}$ [verified]

B4: $h_R^{(1,\text{unconstr})}(0) = \frac{1}{36}$ [verified]

B5: $h_C^{(1)}(x) \to -\frac{1}{3x} + O(x^{-2})$ as $x \to \infty$ [verified]

B6: $h_R^{(1)}(x) \to \frac{1}{9x} + O(x^{-2})$ as $x \to \infty$ [verified]

B7: $h_C^{(1)}(x) = \frac{1}{10} - \frac{11}{420}\,x + O(x^2)$ [verified]

B8: $h_R^{(1)}(x) = \frac{1}{630}\,x + O(x^2)$ [verified]

B9: Ghost subtraction: $h^{(1)} = h^{(1,\text{unconstr})} - 2\,h^{(0)}(\xi{=}0)$ [verified]

B10: No ξ-dependence (gauge invariance fixes coupling) [verified]

B11: $\beta_R^{(1)} = 0$ (conformal invariance of Maxwell) [verified]

B12: $\alpha_{E_4}^{(1)} = -31/180$ (Gauss–Bonnet, topological) [verified]

References

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